Tiago Ferrão-Lindqvist

Test & Validation Engineer (Mechanical) | Python, SQL & Data Analysis

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Professional Summary

Mechanical engineer with 12 years of experience in product design and test & validation, transitioning into data engineering. Develops Python tools for test-data processing, specification checks and automated reporting, reducing report preparation from about three days to half a day per test. Uses SQL Server for trend analysis across product generations. Brings experience translating test requirements into repeatable data checks and engineering decisions.

Technical Skills

Programming and data analysis: Python (pandas, NumPy, Matplotlib), SQL, SQL Server, MATLAB/Simulink

Reporting and analysis tools: Power BI, Excel/VBA, Minitab, design of experiments (DOE)

Engineering tools and methods: ANSYS Mechanical, SolidWorks, CATIA V5, LabVIEW, GD&T, DFMEA, 8D problem solving

Professional Experience

Test & Validation Engineer | Corvane Automotive Components

Mar 2019 – Present

- Developed Python scripts using pandas, NumPy and Matplotlib to read test-rig output files, check every channel against test specifications and generate standard report figures; reduced report preparation from about three days to half a day per test.
- Query the plant's SQL Server test-results database for trend analysis across three product generations.
- Plan and run durability, vibration and climatic tests on seat-adjuster mechanisms across six servo-hydraulic and electric test rigs; prepare test reports for OEM customers.
- Correlate rig results with FEA in ANSYS Mechanical and lead 8D root-cause analysis on field failures; proposed a redesigned adjuster rail bracket that reduced warranty returns for that part by 40%.
- Maintain MATLAB/Simulink models of the adjuster drive to select rig load profiles.

Mechanical Design Engineer | Halvik Pumps

Sep 2014 – Feb 2019

- Designed industrial centrifugal pump casings and impellers in SolidWorks, produced drawings with GD&T and owned DFMEA for two product lines.
- Performed structural FEA in ANSYS; changed one casing family from grey cast iron to ductile iron, reducing material cost by 12%.
- Supported conversion of the pump performance test bench to a Siemens S7 PLC-controlled setup.

Education

MSc Mechanical Engineering | Sado Valley Polytechnic | 2014

Thesis: Fatigue life prediction of welded brackets under variable-amplitude loading

Languages

Portuguese (native) | English (fluent) | Spanish (good)